

Kodre 27/1: Taylor za $\tan(x)$ na 3 načine

① Prek formule

$$y(x) = \tan x \rightarrow y(0) = 0$$

$$y'(x) = \left(\frac{\sin x}{\cos x}\right)' = \frac{1}{\cos^2 x} \rightarrow y'(0) = 1$$

$$y''(x) = +2\cos^{-3} x \sin x \rightarrow y''(0) = 0$$

$$y'''(x) = 2[+3\cos^{-4} x \sin x \sin x + \cos^{-3} x \cos x] \rightarrow y'''(0) = 2$$

$$\rightarrow \underline{\tan x = x + 2\frac{x^3}{3!} + o(x^5)}$$

② Deljenje polinomov

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} + o(x^7)$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} + o(x^6)$$

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots$$

$$\tan x = \frac{\sin x}{\cos x} = \frac{x - \frac{x^3}{3!} + \frac{x^5}{5!} + o(x^7)}{1 - \left(\frac{x^2}{2!} - \frac{x^4}{4!} + o(x^6)\right)} =$$

$$= \left(x - \frac{x^3}{3!} + \frac{x^5}{5!}\right) \left(1 + \left(\frac{x^2}{2!} - \frac{x^4}{4!}\right) + \left(\frac{x^2}{2!} - \frac{x^4}{4!}\right)^2 + o(x^6)\right)$$

$$= x - \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^3}{2!} - \frac{x^5}{3!2!} - \frac{x^5}{4!} + \frac{x^5}{(2!)^2} + o(x^6) =$$

$$\underbrace{\frac{1}{2} - \frac{1}{6} = \frac{2}{6} = \frac{1}{3}} \quad \underbrace{\frac{1}{120} - \frac{1}{12} - \frac{1}{24} + \frac{1}{4} = \frac{1-10-5+30}{120} = \frac{16}{120} = \frac{2}{15} = \frac{2}{15}}$$

$$= x + x^3 \left(\frac{1}{2!} - \frac{1}{3!}\right) + x^5 \left(\frac{1}{5!} - \frac{1}{3!2!} - \frac{1}{4!} + \frac{1}{(2!)^2}\right) + o(x^6) = \underline{x + \frac{x^3}{3} + \frac{2}{15}x^5 + o(x^7)}$$

③ Iteracija (prek inverza, vstavljanje)

$$y = \tan x \rightarrow x = \arctan y = \int_0^y \frac{du}{1+u^2} = \int_0^y [1 - u^2 + u^4 - u^6 + o(u^8)] du$$

$$x = y - \frac{y^3}{3} + \frac{y^5}{5} - \frac{y^7}{7} + o(y^9) \rightarrow y = x + \frac{y^3}{3} - \frac{y^5}{5} + \frac{y^7}{7} + o(y^9)$$

$$y = x + \frac{1}{3} \left[x + \frac{y^3}{3} - \frac{y^5}{5} \right]^3 - \frac{1}{5} \left[x + \frac{y^3}{3} \right]^5 + \frac{1}{7} [x]^7 + o(x^8)$$

$$= x + \frac{1}{3} [x^3 + x^2 y^3 + \frac{x y^6}{3} - \frac{3 x^2 y^5}{5}] - \frac{1}{5} [x^5 + 5 x^4 \frac{y^3}{3}] + \frac{1}{7} x^7 + o(x^8) =$$

$$= x + \frac{1}{3} x^3 - \frac{1}{5} x^5 + \frac{1}{3} x^2 y^3 + \frac{1}{9} x y^6 - \frac{1}{5} x^2 y^5 - \frac{1}{3} x^4 y^3 + \frac{1}{7} x^7 + o(x^8) =$$

$$= x + \frac{1}{3} x^3 - \frac{1}{5} x^5 + \frac{1}{7} x^7 + \frac{1}{3} x^2 \left[x + \frac{y^3}{3} \right]^3 + \frac{1}{9} x [x]^6 - \frac{1}{5} x^2 [x]^5 - \frac{1}{3} x^4 [x]^3 + o(x^8) =$$

$$= x + \frac{1}{3} x^3 - \frac{1}{5} x^5 + \frac{1}{7} x^7 + \frac{1}{3} x^2 [x^3 + x^2 y^3] + \frac{1}{9} x^7 - \frac{1}{5} x^7 - \frac{1}{3} x^7 + o(x^8) =$$

$$= x + \frac{1}{3} x^3 + x^5 \left[-\frac{1}{5} + \frac{1}{3} \right] + x^7 \left[\frac{1}{7} + \frac{1}{9} - \frac{1}{5} - \frac{1}{3} \right] + \frac{1}{3} x^4 [x]^3 + \frac{1}{3} x^3 + \frac{2}{15} x^5 + \frac{17}{5 \cdot 7 \cdot 9} x^7 + o(x^8)$$

$$\frac{45 + 35 - 63}{7 \cdot 9 \cdot 5} = \frac{80 - 63}{7 \cdot 9 \cdot 5} = \frac{17}{7 \cdot 9 \cdot 5}$$

Zbirka, 2.3 Razvij $f(x) = \coth x - \frac{1}{x}$

$$g(x) = \coth x$$

$$g'(x) = -\frac{1}{\sinh^2 x} = -\frac{2}{\cosh 2x - 1} = -\frac{x}{\frac{(2x)^2}{2} \left(1 + \frac{2(2x)^2}{4!} + \frac{2(2x)^4}{6!} + o(x^6) \right)} =$$

$$= -\frac{1}{x^2} \left[1 - \left(\frac{8x^2}{4!} + \frac{32x^4}{6!} \right) + \left(\frac{8x^2}{4!} + \frac{32x^4}{6!} \right)^2 + o(x^6) \right] =$$

$$= -\frac{1}{x^2} \left[1 - \frac{x^2}{3} - \frac{2}{45} x^4 + \frac{x^4}{9} + o(x^6) \right] = -\frac{1}{x^2} + \frac{1}{3} - \frac{1}{45} x^2 + o(x^4)$$

$$\rightarrow g(x) = \frac{1}{x} + \frac{1}{3} x - \frac{1}{45} x^3 + o(x^5) \Rightarrow \underline{f(x) = \frac{x}{3} - \frac{x^3}{45} + o(x^5)}$$

Picardova iteracija

$$g(x) \text{ reši DE } g'(x) = 1 - g^2(x)$$

$$\rightarrow y' = f(t, y(t)) \rightarrow y(t) = y_0 + \int_{t_0}^t f(s, y(s)) ds$$
$$y(t_0) = y_0$$

$$y^{(n+1)}(t) = y_0 + \int_{t_0}^t f(s, y^{(n)}(s)) ds$$

$$g' = 1 - g^2$$

$$g(0) = 0 \text{ izberemo } \rightarrow g^{(1)} = 0 + \int_0^x 1 - \underset{0}{g} dg = x$$

$$g^{(2)} = 0 + \int_0^x (1 - x^2) dx = x - \frac{1}{3} x^3$$

$$g^{(3)} = \int_0^x (1 - [x - \frac{1}{3} x^3]^2) dx = \int_0^x [1 - x^2 + \frac{2}{3} x^4 - \frac{1}{9} x^6] dx = x - \frac{x^3}{3} + \frac{2}{3} \frac{x^5}{5} - \frac{1}{9} \frac{x^7}{7}$$

tanh, ker smo izbrali $g^{(0)}(0) = 0$

$$\rightarrow g^{(0)} = \frac{1}{x}$$

$$g^{(1)} = \int (1 - \frac{1}{x^2}) dx = \underline{x} + x^{-1} + C$$

$$g^{(2)} = \int (1 - [x^2 + 2 + \frac{1}{x^2} + 2xc + \frac{2c}{x} + c^2]) dx = x - \frac{x^3}{3} - 2x + \frac{1}{x} - x^2 c + 2c \ln x + c^2 x + D$$

$$\stackrel{C=D=0}{\rightarrow} + \frac{1}{x} - x - \frac{x^3}{3}$$

$\rightarrow$ Ne konv.

$$\rightarrow g = \frac{h(x)}{x^2} \text{ nova funkc.}$$

$$\rightarrow 1 - \frac{h^2}{x^4} = 1 - g^2 = g' = \frac{h'}{x^2} - 2 \frac{h}{x^3}$$

$$h' = x^2 - \frac{h^2}{x^2} + 2 \frac{h}{x}, h(0) = 0 \rightarrow \text{izkaže se da konv.}$$

Model absorpcije svetlobe $\mu = \mu_0(1 + kj)$; $j(x) = ?$

$$\frac{dj}{dx} = -\mu j \quad \int_0^x dx = \int_{j_0}^j \frac{dj}{-\mu j} = -\frac{1}{\mu_0} \int_{j_0}^j \frac{dj}{j(1+kj)}$$

$$-\mu_0 x = \int_{j_0}^j \frac{dj}{j(1+kj)} = \int_{j_0}^j \frac{dj}{j} - k \int_{j_0}^j \frac{dj}{1+kj} =$$

$$= \ln \frac{j}{j_0} - \ln \frac{1+kj}{1+kj_0} = \ln \frac{j(1+kj_0)}{j_0(1+kj)}$$

$$\frac{A}{j} + \frac{B}{1+kj} = \frac{1}{j(1+kj)}$$

$$A \approx 1$$

$$kA + B = 0$$

$$B = -k$$

$$1 + kj = u$$

$$k dj = du$$

$$\rightarrow \frac{j(1+kj_0)}{j_0(1+kj)} = e^{-\mu_0 x} \rightarrow j(1+kj_0) = j_0 e^{-\mu_0 x} + j j_0 k e^{-\mu_0 x}$$

$$j(1+kj_0(1-e^{-\mu_0 x})) = j_0 e^{-\mu_0 x}$$

$$j = \frac{j_0 e^{-\mu_0 x}}{1+kj_0(1-e^{-\mu_0 x})}$$